

Design FMEA (DFMEA) — Qualitative

Case ID: t3_r1_mimo

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#	Failure Mode	Potential Effect	Potential Cause	Current Controls	Severity
1	Lithium plating during 4C charge	Internal short circuit, capacity loss, safety hazard	Excessive heat generation, insufficient electrolyte safety margin	Enhanced thermal simulation (0.45) + simulation verification	10
2	Thermal runaway during fast charging	Cell venting, fire, explosion	Excessive heat generation	Thermal simulation (T_max 54.5°C < 60°C limit), h=50 W/m²K cooling	9
3	Low 5C capacity retention	Reduced runtime in power tool applications	Excessive limit ion transport	Electrode thinning (35/39 μm) + high porosity design	6
4	SEI growth during cycling	Capacity fade over lifetime	Electrolyte decomposition	Accelerated aging model	Available but not in scope
5	Electrolyte degradation at high temperature	Increased impedance, reduced power	Electrolyte decomposition	EC/EMC ratio 60:40 at 60°C in simulation	6
6	Electrode delamination under high current	Capacity loss, impedance rise	Mechanical stress from rapid insertion	Particle size controlled for stress mitigation	2
7	Separator shutdown failure	Internal short, thermal runaway	Separator melting at elevated temperature	Separator typical shutdown 130°C; T_max well below	10

RPN Scale: Severity (1-10) × Occurrence (1-10) × Detection (1-10). RPN > 100 requires recommended action.

Notes

- This DFMEA is qualitative, based on virtual simulation results and domain knowledge
- Physical validation required for all high-RPN items before production
- The enhanced electrolyte parameters ($\sigma=1.5$, $t_{\text{eff}}=0.45$) are domain estimates — real electrolyte performance must be experimentally verified
- Aging/cycle-life failure modes (SEI growth, capacity fade) were not simulated in this design iteration